

ASSESSMENT-1

CO-1

CSA0904- PROGRAMMING IN JAVA

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```
1. Student management system
class student
Private : name, rollno, marks, average
Method accept Details()
Method calculate Average()
    average = marks/3
Method display-Result()
    If average >= 50
        Display "Pass"
    else
        Display "Fail"
End class

Main
    create student s
    call accept Details()
    call calculate Average()
    call display Result()
```

2. Library Book Management

```
Class Book
Private : title, author, ISBN is Available
Method issue Book()
Method return Book()
Method display Book()

Main
```

create Book b₁, b₂, b₃

Issue b₁

Return b₁

Display books

3. Employee salary calculation

class employee

Private : emp id, name, basic pay

Method calculateGross()

Method calculateNet()

Method display slip()

Main

create Employee e

call calculateGross()

call calculateNet()

call display slip()

4. shape area calculation

class shape

Method calculateArea()

class circle inherits shape

Override calculateArea()

class rectangle inherits shape

Override calculateArea()

class Triangle inherits shape

Override calculateArea()

Main

call calculateArea() for all objects

call calculateArea() for all objects

Vehicle Registration System

class Vehicle

regno, owner, fuelType

Method displayDetails()

class TwoWheeler inherits Vehicle

override displayDetails()

class FourWheeler inherits Vehicle

override displayDetails()

Main

Display details

6. Bank Account Hierarchy

class BankAccount

accNo, balance

Method withdraw()

class SavingsAccount inherits BankAccount

override withdraw()

Method calculateInterest()

class CurrentAccount inherits BankAccount

override withdraw()

Main

Withdraw from both accounts

7. Hospital Patient Record System

class Patient

private: diagnosis, prescription

Method generateBill()

class Inpatient inherits Patient

override generateBill()

Main
Generale bills

8. E-commerce product catalog

class product

Method applyDiscount()

class Electronics inherits product

override applyDiscount()

class clothing inherits product

override applyDiscount()

class Grocery inherits product

override applyDiscount()

Main

Apply discount

9. university staff Hierarchy

class staff

Private : staff / id

protected : salary

Method calculate Allowance()

class Teaching staff inherits staff

class Professor inherits Teaching staff

override calculate Allowance()

class Lecturer inherits Teaching staff

override calculate Allowance()

Main

calculate allowance

Smart Home Device Controller

class SmartDevice

Method executeCommand()

class SmartLight inherits SmartDevice

Override executeCommand()

class SmartThermostat inherits SmartDevice

Override executeCommand()

class: SmartLock inherits SmartDevice

Private: PIN

Override executeCommand()

Main

Execute commands